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Term Paper

Topic 1.14: Checker for correctness of same file- and attribute naming convention using a local LLM

Subject: Forschungspraktikum - Datenmanagement

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Abstract

Modern data integration pipelines usually encounter inconsistencies in file and attribute naming conventions across datasets, resulting in challenges in schema alignment and data quality. Traditional string-based validation techniques are limited in detecting semantic inconsistencies between column names such as “fname”, “first_name”, and “givenName”, resulting in unreliable standardisation.

This study suggests a naming convention checking system using a locally deployed Large Language Model (LLaMA3 via Ollama) to automatically search for dissimilarity and semantically related attributes. The system combines column normalisation, structured prompt engineering, and strict JSON-based output validation to analyse naming patterns and recommend consistent representations.

To evaluate the effectiveness of the approach, the system is also compared with traditional string similarity methods such as Levenshtein Method, Fuzzy Token Sort and Fuzzy Partial using evaluation metrics. The results show that the LLM-based approach produces improved semantic understanding and higher accuracy in identifying inconsistent or equivalent attribute names.

Furthermore, the use of a locally deployed model confirms data privacy and cost efficiency, making the system applicable to data engineering workflows. The findings exhibit the potential of large language models in enforcing naming conventions and improving data consistency in heterogeneous datasets.

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1 Introduction

1.1 Problem Statement

Traditional methods for validating naming conventions primarily utilize string similarity measures or rule-based techniques. Approaches such as Levenshtein distance and token-based matching can detect syntactic similarity but do not capture semantic relationships between attributes. Consequently, these methods frequently fail to identify semantically equivalent or inconsistent attribute names, particularly when abbreviations, domain-specific terminology, or diverse naming styles are present. This limitation leads to incomplete or inaccurate validation of naming conventions, increasing the risk of data inconsistency and reducing the reliability of data integration processes.

Despite significant advancements in data integration techniques, the problem of semantic inconsistency in attribute naming remains largely unsolved, particularly in heterogeneous and unstructured datasets. Traditional approaches fail to capture contextual meaning, leading to incorrect or incomplete schema alignment. This motivates the exploration of semantic methods using large language models, which have demonstrated strong capabilities in understanding contextual relationships between textual entities.

1.2 Research Objective

The main objective of this study is to develop a system that can automatically check and validate file and attribute naming conventions using a locally deployed Large Language Model. The specific objectives include:

1. To design a system capable of identifying semantically equivalent and inconsistent attribute names across datasets
2. To leverage a local LLM (LLaMA3 via Ollama) for context-aware semantic analysis
3. To generate structured and interpretable outputs for naming validation
4. To compare the proposed approach with traditional string similarity methods

1.3 Hypothesis

This study is based on the following hypothesis:

H1: A locally deployed Large Language Model can more accurately identify semantic inconsistencies in a file compared to traditional string-based methods.

H0: There is no significant difference between LLM-based and traditional methods in identifying naming inconsistencies.

2 State of the Art

2.1 Limitations of Traditional Methods

Existing approaches for validating file and attribute naming conventions primarily rely on string-based similarity measures and rule-based techniques. Common methods such as Levenshtein distance, token-based matching, and fuzzy string matching compare textual similarity at the character or token level. These techniques are effective in detecting minor variations such as spelling errors or formatting differences.

However, traditional methods are limited in capturing semantic relationships between attribute names. Schema matching is defined as the task of identifying semantic correspondences between elements of different schemas, which is inherently challenging due to variations in naming conventions [1]. Attributes representing the same concept may have completely different names, while similar names may refer to different concepts, making purely string-based comparison unreliable [1].

Furthermore, studies highlight that traditional methods rely heavily on syntactic similarity and predefined rules, without considering contextual meaning or domain knowledge [2]. As a result, these techniques often fail to detect semantically equivalent attributes such as *"fname"* and *"first_name"*, leading to inaccurate or incomplete validation of naming conventions.

2.2 Research Gap

Despite advancements in schema matching techniques, accurately identifying semantic inconsistencies in attribute naming remains a challenging problem. Existing approaches either rely on syntactic similarity or require complex feature engineering and labeled datasets, limiting their scalability and generalization [3].

Recent developments in Large Language Models (LLMs) provide improved semantic understanding and reasoning capabilities. LLMs have demonstrated strong performance in tasks involving natural language understanding and can identify semantic relationships between textual inputs [4], [5]. This makes them suitable for detecting inconsistencies and equivalences in attribute naming.

However, the application of LLMs for automated naming convention validation, particularly in local and privacy-preserving environments, is still limited. Most existing solutions depend on cloud-based models, raising concerns related to data privacy and cost. This highlights a clear research gap in developing efficient, secure, and semantically aware systems for attribute name standardisation.

3 System Architecture

3.1 Use Case Diagram

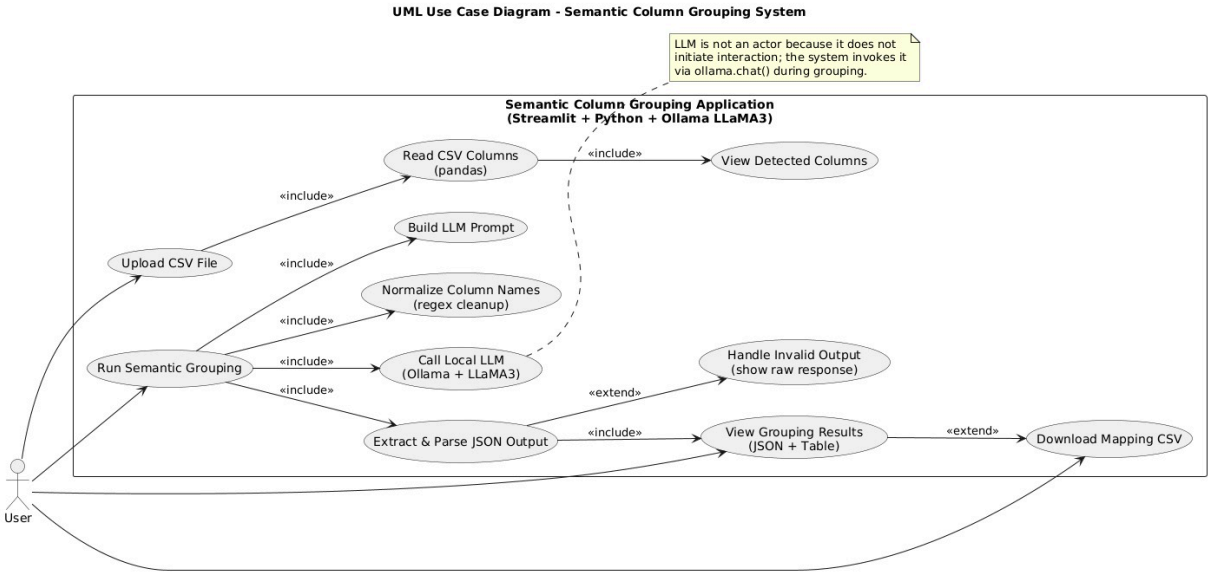

Figure 3.1.1: UML Use Case Diagram of the Semantic Grouping System

The diagram illustrates the interaction between the user and the system, highlighting the main use cases and their relationships in the semantic column grouping process.

3.2 Data Flow Diagram

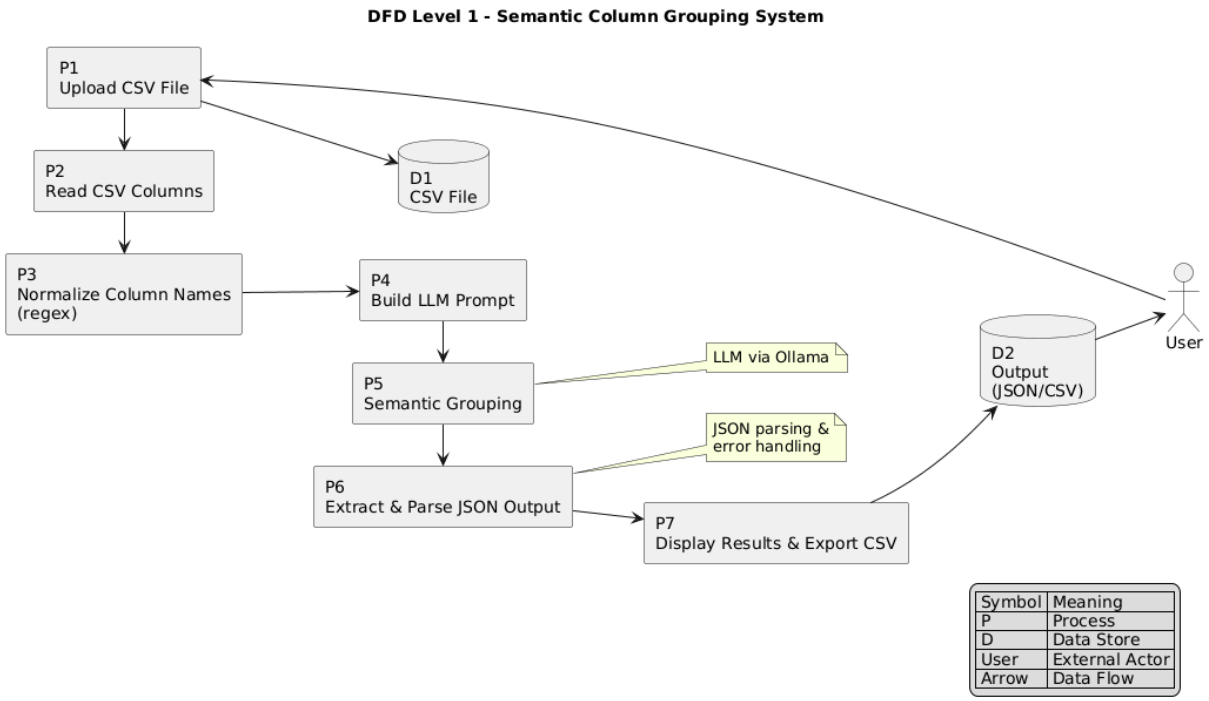

Figure 3.2.1: DFD Level 1 of Semantic Column Grouping System

This diagram represents the data flow from input CSV processing to LLM-based semantic grouping and output generation.

3.3 System Architecture Diagram

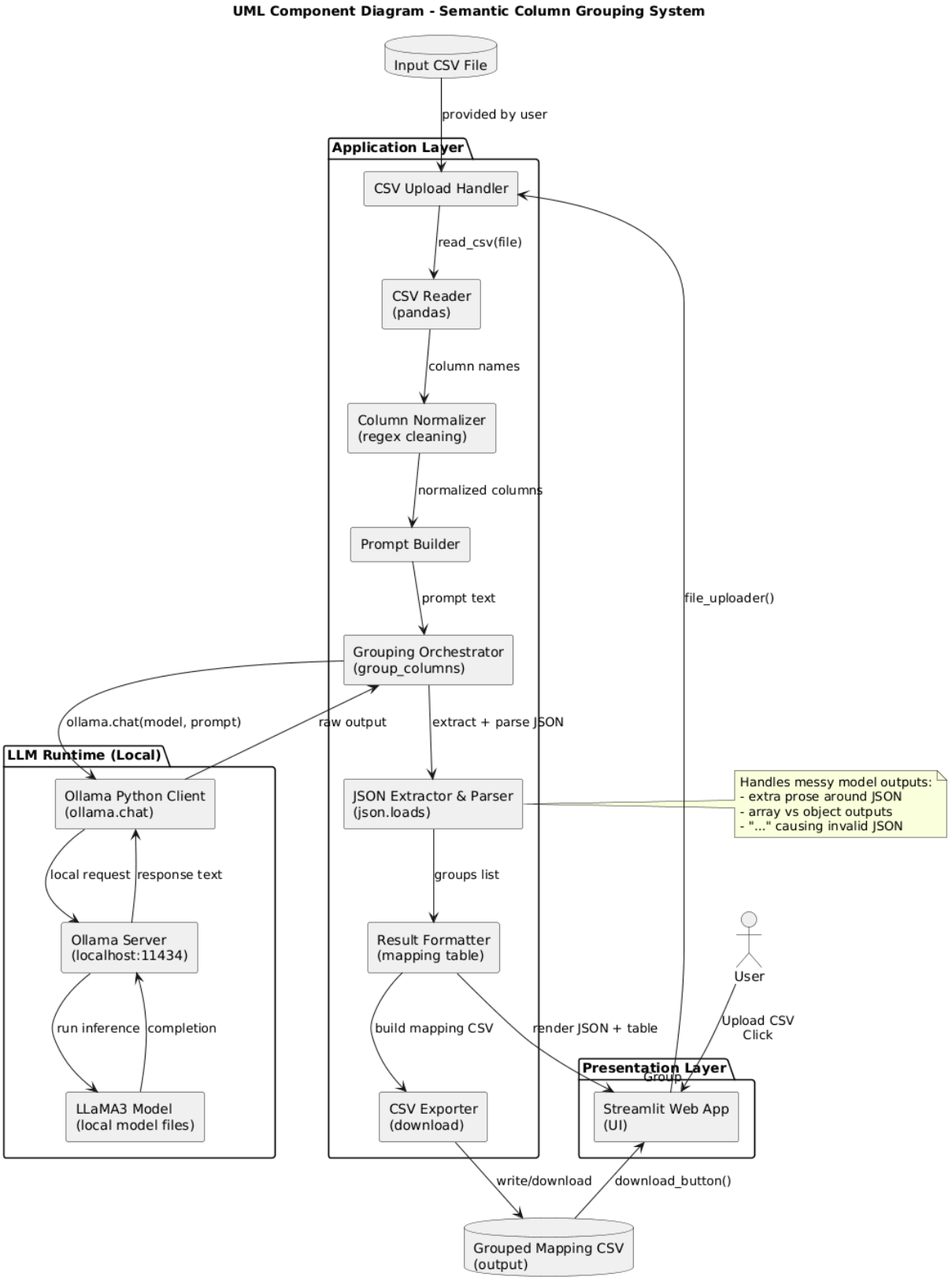

Figure 3.3.1: UML Component Diagram of Semantic Column Grouping System

This diagram represents the logical architecture of the system, illustrating the flow of data through different components including CSV processing, column normalization, prompt generation, LLM-based semantic grouping, and JSON parsing. It also highlights the interaction with the local LLM runtime (Ollama) and the presentation layer for displaying results to the user.

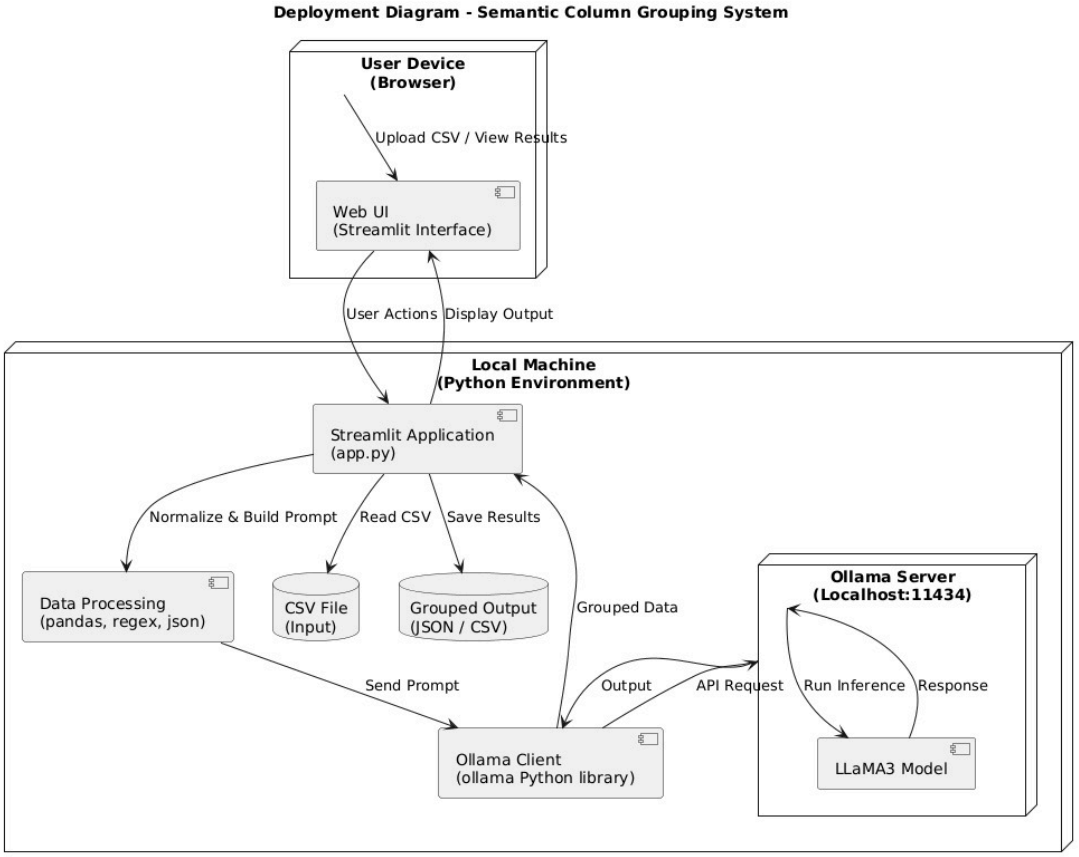

Figure 3.3.2 : Deployment Diagram of Semantic Column Grouping System

This diagram illustrates the physical deployment of the system, showing how the Streamlit application runs on a local machine and interacts with the Ollama server hosting the LLaMA3 model. It also depicts data flow between the user interface, processing modules, and the LLM runtime for generating grouped column outputs.

4 Methodology

4.1 Problem Definition

The problem addressed in this study is the automatic validation and standardisation of file and attribute naming conventions in heterogeneous datasets. Given a set of column names, the objective is to identify semantically equivalent attributes, detect inconsistencies, and group them under a consistent representation. Formally, the task can be defined as partitioning a set of attributes into clusters such that each cluster represents a single semantic concept, while avoiding incorrect grouping of attributes belonging to different entities. This formulation enables systematic evaluation of the proposed approach.

4.2 System Workflow

The proposed system follows a structured workflow to perform semantic analysis of attribute names. Initially, a CSV file is provided as input, from which column names are extracted. These column names are then preprocessed through normalization steps including lowercasing, removal of special characters, and splitting of compound words. A structured prompt is generated and sent to a locally deployed Large Language Model (LLaMA3 via Ollama), which analyses the semantic relationships between attributes and groups them accordingly. The model output is then processed and presented to the user in a structured format, with an option to export the results.

4.3 Prompt Design

Prompt engineering is a key component of the proposed approach, as it guides the behaviour of the Large Language Model. The prompt is designed to define the model's role as a semantic analyzer and includes explicit constraints to ensure accurate results. These constraints include grouping only semantically equivalent attributes, avoiding merging of different entity types, and producing output strictly in a predefined JSON format. By enforcing these rules, the system ensures consistency, reduces ambiguity, and improves the reliability of the generated output.

4.4 Output Processing

The output generated by the LLM is parsed to extract structured JSON data representing grouped attributes. Since LLM responses may include additional text or formatting inconsistencies, the system incorporates validation mechanisms to ensure that the output conforms to the expected structure. Invalid elements are removed, and missing or incorrect fields are handled appropriately. The final validated output includes grouped attributes, recommended naming conventions, and confidence scores, which are displayed to the user and can be exported for further use. The confidence score indicates the model's level of certainty for each grouping and is used as an additional reference to assess the reliability of the results. This score is generated internally by the language model based on its predictions, and its exact calculation is not explicitly defined or controlled within the system.

5 Evaluation

5.1 Dataset Description

The evaluation of the proposed naming convention checking system was conducted using multiple CSV datasets of varying sizes and complexity. The datasets range from small samples to larger datasets containing over 100 columns and more than 100 rows, enabling comprehensive testing across different scenarios. The column names were intentionally designed to include semantically similar attributes with increasing levels of difficulty, such as variations like *"fname"*, *"first_name"*, and *"firstname"*, which may appear at different positions within the dataset to simulate real-world inconsistencies. Additionally, controlled typographical variations were introduced to further challenge the system, including misspelled or noisy attribute names such as *"frist_name"*, *"fistname"*, and *"nmae"*. These variations are intended to simulate real-world data quality issues and to evaluate the robustness of the system in handling noisy and unstructured inputs. This diversity in dataset design ensures that the evaluation effectively assesses the system's ability to identify semantic equivalence, handle heterogeneous naming styles, and detect inconsistencies across complex data environments.

5.2 Baseline Methods and Metrics

To evaluate the performance of the proposed system, it is compared with traditional string similarity methods that are commonly used for textual matching tasks. These methods rely on syntactic similarity rather than semantic understanding and serve as baseline approaches for comparison. Levenshtein Distance is a widely used metric that calculates the minimum number of character edits required to transform one string into another, making it effective for detecting typographical differences but not semantic relationships. Fuzzy Token Sort Ratio compares strings by sorting their tokens and measuring similarity, which helps handle variations in word order but still depends on textual similarity. Similarly, Fuzzy Partial Ratio measures similarity by identifying the best matching substring between two strings, making it useful when one string is a subset of another but insufficient for capturing semantic equivalence. Although these methods are effective for detecting syntactic similarity, they fail to identify semantically equivalent attribute names such as "fname" and "first_name". Therefore, they are used as baseline methods to evaluate the effectiveness of the proposed LLM-based approach. The evaluation is conducted using standard metrics such as precision and recall, where precision measures the proportion of correctly identified attribute groupings among all predicted groupings, and recall measures the proportion of actual correct groupings that are successfully identified.

5.3 Experimental Setup and Results

The evaluation of the proposed system was conducted to assess its effectiveness in handling diverse and complex naming conventions. The datasets included variations such as abbreviations, inconsistent formatting, and controlled typographical errors, with semantically equivalent attributes placed at different positions to increase the difficulty of the grouping task. The performance of the system was compared with traditional string similarity methods, including Levenshtein Distance, Fuzzy Token Sort, and Fuzzy Partial Ratio. Each method was applied to the same datasets, and the resulting groupings were evaluated based on their correctness using precision and recall metrics. The results demonstrate that the proposed LLM-based approach provides improved accuracy in identifying semantically equivalent attributes compared to traditional syntactic methods. As shown in Figure 5.3.1, the LLM-based method achieves higher recall compared to baseline methods.

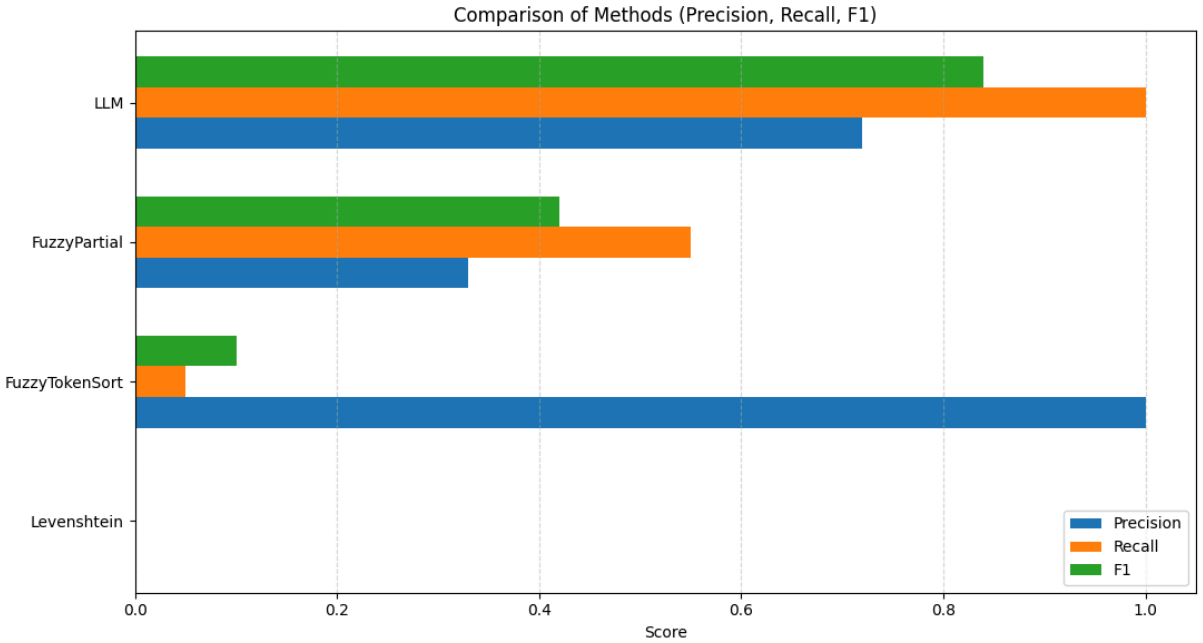

Figure 5.3.1

The figure represents the performance comparison of traditional and LLM-based methods, highlighting improvements in precision, recall, and F1-score.

For evaluation, a pairwise comparison approach was used, where each possible pair of attribute names is classified as either matching or non-matching based on the ground truth. The performance of each method was then measured using precision, recall, and F1-score. Precision represents the proportion of correctly identified matches, recall represents the proportion of actual matches detected, and F1-score provides a balanced measure of both.

6 Results and Discussion

6.1 Analysis and Comparison

The evaluation results demonstrate a clear difference between syntactic and semantic approaches for attribute name matching. Traditional string similarity methods such as Levenshtein Distance, Fuzzy Token Sort, and Fuzzy Partial Ratio rely on textual comparison and are effective in identifying minor variations such as spelling differences or changes in word order, for example matching *"first name"* and *"name first"*. However, these methods struggle when attribute names differ significantly in structure due to abbreviations, formatting variations, or typographical errors, as seen in examples like *"f_name"*, *"first_name"*, and *"frstname"*, where syntactic similarity is low despite representing the same concept. In contrast, the proposed LLM-based approach performs semantic analysis by understanding the meaning and context of attribute names, allowing it to correctly identify such variations as semantically equivalent. This results in improved precision and recall, demonstrating the ability of the model to handle noisy and heterogeneous data more effectively. Overall, the comparison highlights that while syntactic methods are suitable for simple matching tasks, they are limited in real-world scenarios, whereas the LLM-based approach provides a more robust and accurate solution for naming convention validation.

6.2 Validation of Hypothesis

The hypothesis that a locally deployed Large Language Model can improve the accuracy of naming convention validation compared to traditional string similarity methods is supported by the experimental results. The LLM-based approach achieves higher precision, recall, and F1-score, indicating its ability to accurately identify semantically equivalent attributes. Although no formal statistical significance testing is performed, the consistent improvement across multiple datasets suggests that the proposed approach provides a more effective solution for semantic attribute grouping.

The improved precision and recall values demonstrate that the LLM-based approach is better able to capture the meaning and context of attribute names, allowing it to correctly group semantically equivalent attributes even when they differ in structure, formatting, or contain typographical errors. In contrast, syntactic methods rely solely on textual similarity and often fail in such scenarios. The consistent performance of the proposed system across datasets of varying complexity further validates its effectiveness. Therefore, the findings confirm that semantic analysis using a locally deployed LLM provides a more accurate and robust solution for naming convention validation, supporting the proposed hypothesis.

7 Limitations

1. The performance of the proposed system is highly dependent on prompt design, and variations in prompt structure can lead to inconsistent or non-structured outputs.
2. The use of a locally deployed Large Language Model requires higher computational resources compared to traditional string similarity methods, which may affect scalability for large datasets.
3. The system relies only on attribute names and does not consider data values or domain-specific information, which could further improve accuracy.
4. The LLM may occasionally produce ambiguous or partially correct groupings, requiring additional validation and post-processing.
5. The LLM may produce non-deterministic outputs, resulting in variations across different runs.

8 Future Scope

Future work will focus on integrating embedding-based similarity measures with LLM outputs to further improve accuracy. Additionally, incorporating data values alongside attribute names could provide richer context for semantic analysis. The development of benchmark datasets for schema matching tasks and evaluation on large-scale real-world datasets also represents an important direction for future research.

9 Conclusion

The findings of this study demonstrate the potential of large language models for addressing challenges in schema matching and naming convention validation. By leveraging semantic understanding, the proposed approach overcomes the limitations of traditional syntactic methods and provides a more accurate and robust solution. The use of a locally deployed model further ensures data privacy and cost efficiency, making the system suitable for practical applications. Overall, this work highlights the effectiveness of semantic analysis and opens new directions for research in data integration and schema alignment.

10 References

[1] E. Rahm and P. A. Bernstein, "A Survey of Approaches to Automatic Schema Matching," VLDB Journal, vol. 10, no. 4, pp. 334–350, 2001.

[2] A. K. Elmagarmid, P. G. Ipeirotis, and V. S. Verykios, "Duplicate Record Detection: A Survey," IEEE Transactions on Knowledge and Data Engineering, vol. 19, no. 1, pp. 1–16, 2007.

[3] P. Christen, "Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection," Springer, 2012.

[4] T. Brown et al., "Language Models are Few-Shot Learners," Advances in Neural Information Processing Systems (NeurIPS), 2020.

[5] H. Touvron et al., "LLaMA: Open and Efficient Foundation Language Models," arXiv preprint arXiv:2302.13971, 2023.

11 Appendix

11.1 Project Repository

The complete implementation of the proposed system, including datasets, source code, and outputs, is available at:

https://github.com/davemeeta/Attribute-Equivalency-Checker.git

11.2 Sample Datasets

A sample dataset used for evaluation is included in the repository to demonstrate the variation in naming conventions.

11.3 Outputs

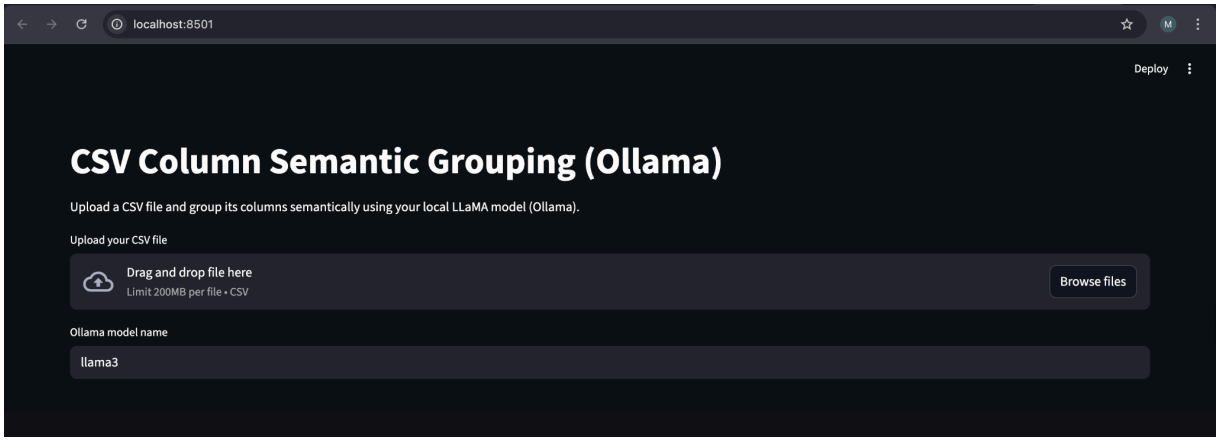

Figure 11.3.1: User interface of the Semantic Grouping system before file upload

Figure 11.3.2: Interface with selected dataset for semantic grouping

Figure 11.3.3: Display of detected column names extracted from the uploaded CSV file

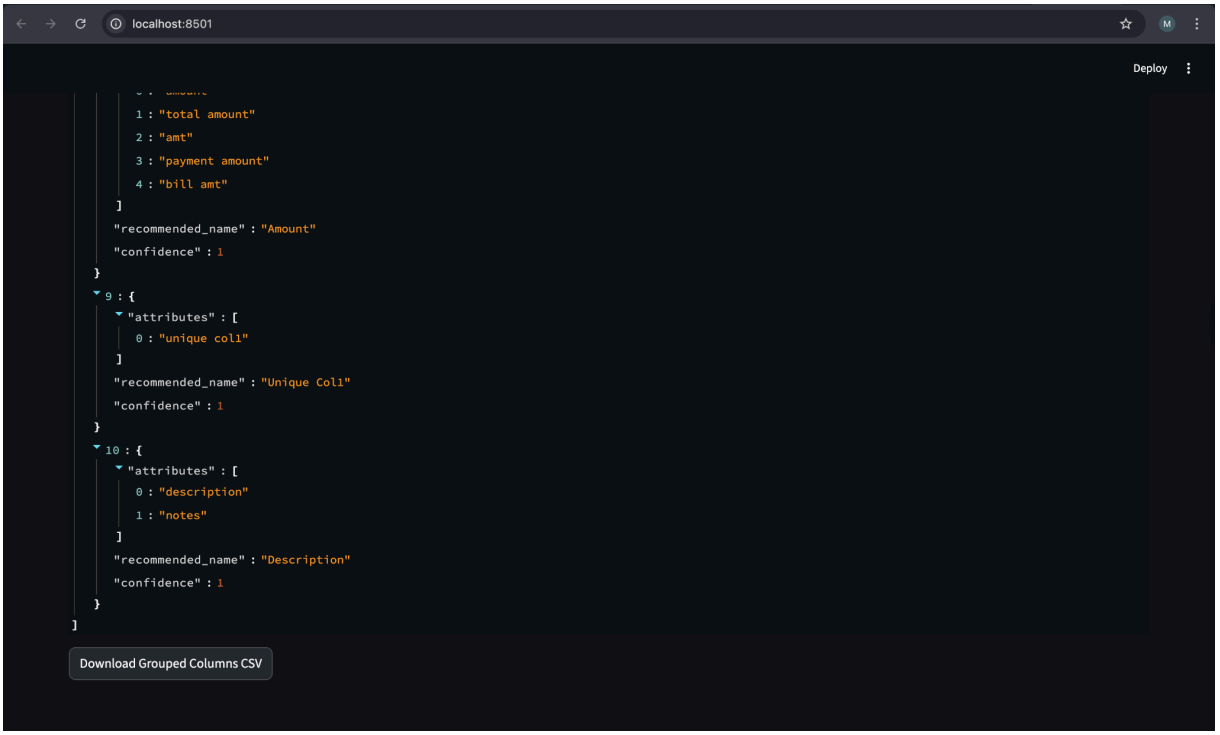

Figure 11.3.4 : Final grouped output with option to download results as a CSV file